

4.5.19. Prove directly from the definitions that for all integers  $m$  and  $n$ , if  $m$  and  $n$  have the same parity, then  $5m + 7n$  is even. Divide into two cases:  $m$  and  $n$  are both even and  $m$  and  $n$  are both odd.

Suppose that  $m$  and  $n$  are integers.

We must show that for all  $m$  and  $n$ , if  $m$  and  $n$  have the same parity, then  $5m + 7n$  is even.

Case 1:  $m$  and  $n$  are even

Definition of even:

$n$  is even  $\leftrightarrow n = 2k$ , for some  $k \in \mathbb{Z}$

Let  $m, n, k, j, i \in \mathbb{Z}$

$$m = 2j$$

$$n = 2i$$

$$5m = 10j$$

$$7n = 14i$$

$$10j + 14i = 2(5j + 7i)$$

$$k = 5j + 7i$$

$$5m + 7n = 2k$$

$2k$  fulfills the definition of even,

for some  $k \in \mathbb{Z}$ .

Therefore,  $5m + 7n$  is even.

Case 2:  $m$  and  $n$  are odd

Definition of odd:

$n$  is odd  $\leftrightarrow n = 2k + 1$ , for some  $k \in \mathbb{Z}$

Let  $m, n, k, j, i \in \mathbb{Z}$

$$m = 2j + 1$$

$$n = 2i + 1$$

$$5m = 5(2j + 1) = 10j + 5$$

$$7n = 7(2i + 1) = 14i + 7$$

$$10j + 5 + 14i + 7 = 10j + 14i + 12 = 2(5j + 7i + 6)$$

$$k = 5j + 7i + 6$$

$$5m + 7n = 2k$$

$2k$  fulfills the definition of even,

for some  $k \in \mathbb{Z}$ .

Therefore,  $5m + 7n$  is even.

Solution: If  $m$  and  $n$  have the same parity, then  $5m + 7n$  is even.